

Diagonal Fourier extension for positively curved surfaces in three dimensions

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Abstract

We prove the diagonal Fourier extension conjecture for compact smooth positively curved surfaces $\Sigma \subset \mathbb{R}^3$, including surfaces with smooth boundary. For every $3 < p < \infty$, the extension operator maps $L^p(\Sigma)$ boundedly into $L^p(\mathbb{R}^3)$. The compact paraboloid case also yields free Schrödinger local smoothing in two spatial dimensions for every $3 < p < \infty$ and Sobolev order $s > 2 - 6/p$.

1 Introduction

Let $\Sigma \subset \mathbb{R}^3$ be a compact smooth surface, with induced surface measure $d\sigma$. Its Fourier extension operator is

$$E_\Sigma f(x) = \int_\Sigma f(\xi) e^{2\pi i x \cdot \xi} d\sigma(\xi).$$

We allow smooth boundary. Positive curvature means that the second fundamental form is definite at every point, with the normal chosen on each chart so that the form is positive definite.

Theorem 1.1. *Let $\Sigma \subset \mathbb{R}^3$ be a compact smooth positively curved surface, possibly with smooth boundary. For every $3 < p < \infty$, there is a finite constant $C_{\Sigma,p}$ such that*

$$\|E_\Sigma f\|_{L^p(\mathbb{R}^3)} \leq C_{\Sigma,p} \|f\|_{L^p(\Sigma)} \tag{1.1}$$

for every complex-valued $f \in L^p(\Sigma)$.

The theorem establishes the diagonal extension conjecture in three dimensions for positively curved surfaces. It includes the sphere, compact elliptic paraboloid patches, and general smooth elliptic graphs. The surface and the exponent are fixed throughout each estimate.

Background. The restriction problem originates in Stein’s work; the Tomas–Stein theory gives $L^2(\Sigma) \rightarrow L^4(\mathbb{R}^3)$ for positively curved surfaces [15, 19]. The bilinear framework of Tao, Vargas, and Vega [18], Wolff’s cone estimate [22], and Tao’s paraboloid estimate [17] developed the role of wave packets and transverse interactions. Bennett, Carbery, and Tao’s multilinear estimates [1] and Bourgain–Guth’s approach [3] further connected these interactions to the linear problem.

Guth’s polynomial-partitioning argument established the bounded-data estimate for $p > 13/4$ [7, Theorem 0.1]. Shayya obtained diagonal estimates in that range [14, Corollary 2.1(iii)]; see also Kim’s treatment [8, Theorem 1.2 and the following remark]. Wang’s broom method further improved bounded-data restriction for the truncated paraboloid [20, Theorem 1.4]. Wang and Wu proved

the diagonal bound for $p > 22/7$ on compact positively curved surfaces, including surfaces with boundary, using decoupling and two-ends Furstenberg inequalities [21, Theorem 0.2].

For the sphere and quadratic graphs, factorization gives a separate route from bounded-data estimates to diagonal estimates with an arbitrarily small increase in the exponent; see Bourgain [2] and Buschenhenke [4, Theorem 1.1 and Remark 1.6]. The argument here treats general smooth elliptic charts through the packet theorem's full array formulation and the explicit phase continuation. The packet and maximal inputs play distinct roles: the former controls oscillatory coefficients, while the latter controls their positive tube density.

The compact paraboloid case also gives a consequence for free Schrödinger evolution. Let $W^{s,p}(\mathbb{R}^2)$ denote the inhomogeneous Bessel-potential space with norm $\|f\|_{W^{s,p}} = \|(I - \Delta)^{s/2} f\|_{L^p(\mathbb{R}^2)}$.

Corollary 1.2 (Free Schrödinger local smoothing). *For every $3 < p < \infty$ and $s > 2 - 6/p$, the free Schrödinger propagator, initially defined on Schwartz functions, extends boundedly from $W^{s,p}(\mathbb{R}^2)$ to $L^p(\mathbb{R}^2 \times [0, 1])$:*

$$\|e^{it\Delta} f\|_{L^p(\mathbb{R}^2 \times [0, 1])} \leq C_{p,s} \|f\|_{W^{s,p}(\mathbb{R}^2)}. \quad (1.2)$$

The spacetime norm is global in space and local in time. Rogers's necessary condition [13, Section 2] shows that the Sobolev threshold cannot be lowered. We deduce the corollary from the compact-frequency transfer of Lee, Rogers, and Seeger [10, Theorem 1.1] in Section 7.

A second consequence concerns local oscillatory integrals with phase $N[x \cdot y + \psi(t; y)]$, where $x, y \in \mathbb{R}^2$, $t \in \mathbb{R}$, and the amplitude is smooth and compactly supported. Suppose $D_y^2 \partial_t \psi$ is definite and its time derivative is a scalar multiple of itself. A change of output variables then reduces the phase to a fixed elliptic graph. Corollary 7.1 gives the resulting L^p estimate with the scaling factor $N^{-3/p}$ for every finite $p > 3$. The additional phase condition is the translation-invariant form of Bourgain's condition; we explain the straightening directly in Section 7, following the setting of Gao–Liu–Xi [5].

The proof of Theorem 1.1 uses two theorem inputs. The packet propagation theorem [11, Theorem 6.4] controls arbitrary finite complex coefficient arrays through an elliptic capacity, with coordinate masks permitted. The Takeya maximal theorem [12, Theorem 1.1] controls nonnegative weighted tube concentration. We state both inputs in the forms used here and verify their hypotheses. The remainder of the proof gives the density conversion, the estimates for rough L^3 graph data, and the passage to general surfaces.

The argument. After fixing a graph chart, an explicit continuation places its phase in the class allowed by the packet theorem. The construction matches the quadratic Taylor data in a transition annulus of a continued spherical patch and preserves uniform ellipticity. It applies equally to boundary charts, where the graph extends smoothly and the datum is extended by zero.

The packet theorem gives a cubic sample estimate with a small spatial power loss and an elliptic-capacity factor. A decomposition according to the capacity of the remaining coefficients allows this factor to be summed using the positive weighted tube estimate. The latter follows from the Takeya maximal theorem by duality; its direction and position formulation follows the classical weighted tube framework of Tao, Vargas, and Vega [18, arXiv version 1, Section 3].

To pass to L^3 data, we prove one joint estimate for a sparse family of balls. Removing a small union of frequency resonances makes the corresponding packet analyses almost orthogonal. The bound is uniform in the ball locations and in arbitrarily large separations. A covering of each level set then removes the spatial loss. This follows Tao's sparse-ball strategy for epsilon removal [16]; see the detailed treatments in Bourgain–Guth's appendix [3] and Kim's appendix [8]. We prove the precise covering and summation needed for the present estimate.

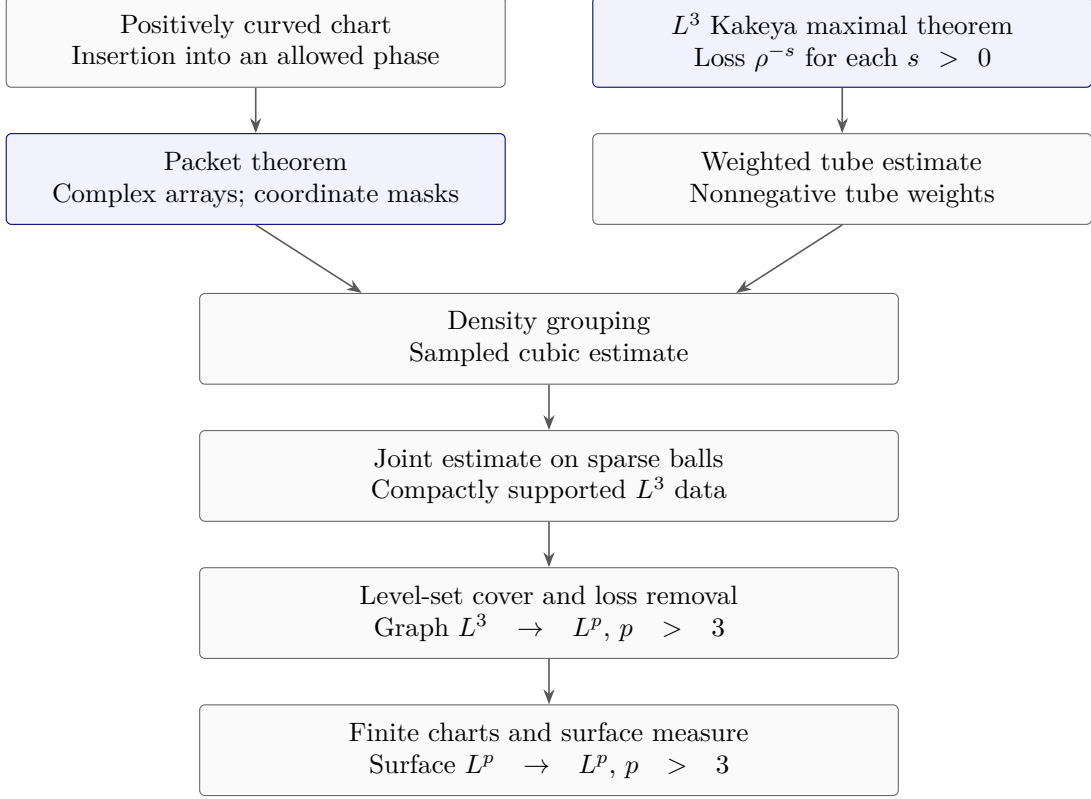


Figure 1: The two independent inputs meet in the sampled cubic estimate. The sparse-ball bound then supplies the common L^3 control needed for globalization and passage to the surface.

The resulting graph bound has L^3 input and L^p output for every $p > 3$. A finite chart decomposition, with the surface and physical-space Jacobians retained, gives Equation (1.1) by the finite-measure embedding $L^p(\Sigma) \subset L^3(\Sigma)$.

Figure 1 records the two inputs and the successive reductions.

Dependence on the exponent. Let $C_{\Sigma,p}^{\text{best}}$ denote the operator norm in Equation (1.1). Write $M(s) \geq 1$ for the constant in the Kakeya maximal estimate with loss δ^{-s} , and let $P_\Sigma(\kappa, \eta) \geq 1$ be the largest packet constant among the finitely many fixed chart and window setups used below. For $3 < p \leq 4$, put $\theta = 10^{-6}(p-3)^2$. The proof gives

$$C_{\Sigma,p}^{\text{best}} \leq C(\Sigma)(p-3)^{-7/p} P_\Sigma(\theta, \theta)^{1/p} M(\theta)^{3/(2p)}, \quad (1.3)$$

where $C(\Sigma)$ is independent of $p \in (3, 4]$. The two input theorems supply finite constants at each positive loss exponent. No rate for their growth is assumed, so Equation (1.3) records the elementary losses without asserting a polynomial bound in $(p-3)^{-1}$. For nonempty Σ , a fixed smooth datum also gives

$$C_{\Sigma,p}^{\text{best}} \geq c(\Sigma)(p-3)^{-1/p}, \quad 3 < p \leq 4. \quad (1.4)$$

Proposition 6.5 proves this comparison.

Conventions and organization. Write $e(t) = \exp(2\pi it)$. All functions and coefficient arrays may be complex-valued. Constants may depend on the fixed surface, charts, frequency supports,

and windows; dependence on the small loss exponents is displayed through P and M . Section 2 fixes the local graph problem. The next sections give the packet interface, the density conversion, and the joint sparse-ball estimate. The global argument tracks the dependence on p , and Section 7 completes the transfer to Σ and proves the local-smoothing and oscillatory-integral consequences.

2 Local graphs and fixed parameters

We first explain the class of graph integrals to be estimated. Near each point of Σ , choose tangent coordinates and a normal in which the surface is a graph with positive definite Hessian. At a boundary point, a smooth parametrization up to the boundary extends to an open neighborhood of the parameter half-disk; this is the usual smooth-boundary convention [9, Chapter 1, pp. 27–28]. Projection onto the tangent plane has invertible differential there. The inverse function theorem on the extension therefore gives a graph on an open planar neighborhood, whose restriction parametrizes the original surface piece.

After shrinking a chart, Lemma 3.1 gives a fixed invertible affine change of ambient frequency coordinates in which the graph agrees locally with a phase $\phi \in C^\infty(\mathbb{R}^2)$ in the precise continuation class of Theorem 3.4. In particular, this phase satisfies

$$cI \leq D^2\phi \leq CI, \quad \sup_{\mathbb{R}^2} |D^j\phi| < \infty \quad (j \geq 2). \quad (2.1)$$

The elementary arguments below use only Equation (2.1); the packet estimate also uses the continuation form verified in Lemma 3.1.

Fix a smooth nonnegative function χ supported in a square centered at zero, such that

$$\sum_{\nu \in a\mathbb{Z}^2} (\chi_a^\nu(\xi))^2 = 1, \quad \chi_a^\nu(\xi) = \chi((\xi - \nu)/a), \quad a > 0. \quad (2.2)$$

For example, start with a compactly supported bump positive on $[-1, 1]^2$, and divide it by the square root of the sum of the squares of its integer translates. This denominator is smooth, positive, and periodic. Let $D_f \geq 1$ be a fixed side length for a square containing the support of χ .

Write

$$E_\phi g(y) = \int_{\mathbb{R}^2} g(\xi) e(y' \cdot \xi + y_3 \phi(\xi)) d\xi, \quad y = (y', y_3) \in \mathbb{R}^2 \times \mathbb{R}.$$

A partition at scale one reduces each compactly supported graph integral to finitely many functions of the form

$$F(y) = E_\phi(h\chi_1^{\nu_*})(y). \quad (2.3)$$

Here $\nu_* \in \mathbb{Z}^2$ is fixed and $h \in L^3(\mathbb{R}^2)$ is supported in a fixed compact set Q_0 . To see the reduction, write graph data $g = \sum_{\nu_*} (g\chi_1^{\nu_*})\chi_1^{\nu_*}$, using Equation (2.2); only finitely many terms meet $\text{supp } g$. The extra copy of the window is included in h .

All constants for a fixed setup may depend on ϕ, Q_0, χ, D_f , and ν_* . These objects, and all label ranges used later, are chosen before the large scale or the small loss exponents vary. In particular,

$$(\nabla\phi(\xi) - \nabla\phi(\eta)) \cdot (\xi - \eta) \geq c|\xi - \eta|^2 \quad (2.4)$$

on $\mathbb{R}^2$. Thus the velocity map $U_\xi = -\nabla\phi(\xi)$ is bi-Lipschitz on each fixed bounded region and has bounded size there.

The final graph estimate controls Equation (2.3) by $\|h\|_3$. We record the precise surface Jacobians in Section 7.

3 The packet estimate and its phase class

We need a cubic bound for the samples obtained by synthesizing a finite packet array and analyzing it at a larger scale. The input array must remain arbitrary, since Section 4 splits it into coordinate subarrays. We first verify the precise phase requirement of the cited packet theorem: uniform convexity alone is not its stated hypothesis.

3.1 Continued phases and local changes of coordinates

The setup of [11, Section 6.1] starts with a small spherical graph and extends its phase smoothly to $\mathbb{R}^2$, with uniformly definite bounded Hessian and bounded derivatives of every order at least two. Its constants may depend on this fixed continuation, the fixed bounded frequency region, and the windows. The continuation is made by cutting off the difference from the quadratic Taylor polynomial. We shall use continuations of precisely this form: write

$$q_s(\xi) = -\sqrt{1 - |\xi|^2}, \quad Q_s(\xi) = -1 + \frac{1}{2}|\xi|^2,$$

and, for a sufficiently small fixed $\rho > 0$, set

$$\phi = Q_s + \zeta(q_s - Q_s), \quad 0 \leq \zeta \leq 1, \quad \zeta = 1 \text{ on } B(0, \rho), \quad \text{supp } \zeta \subset B(0, 2\rho). \quad (3.1)$$

Here ζ is smooth, the product is extended by zero outside $B(0, 2\rho)$, and the resulting phase must satisfy Equation (2.1). No radially is required of the cutoff.

Lemma 3.1 (Insertion of an elliptic graph). *Every germ of a smooth graph with positive definite Hessian can be carried by an invertible affine transformation of $\mathbb{R}^3$ to a graph agreeing, on an open neighborhood, with a phase of the form in Equation (3.1). The phase, the neighborhood, and the affine transformation can all be fixed independently of the packet scales and of the exponents used below.*

Proof. Choose a smooth cutoff ζ_0 equal to one on $B(0, \rho)$, supported in $B(0, 2\rho)$, and identically $1/2$ on a neighborhood V of a point ξ_1 with $\rho < |\xi_1| < 2\rho$. It can be chosen by rescaling a fixed cutoff with an annular plateau. The phase

$$q = Q_s + \zeta_0(q_s - Q_s)$$

has Hessian $I + O(\rho^2)$: the remainder $q_s - Q_s$ and its derivatives of orders zero, one, and two are respectively $O(\rho^4)$, $O(\rho^3)$, and $O(\rho^2)$ on the support of the cutoff. Fix ρ small enough that q'' is positive definite everywhere.

After translating the target base point, let its height function be h_0 near zero and put $H = h_0''(0) > 0$ and $B = q''(\xi_1) > 0$. Choose an invertible real matrix A with $A^\top H A = B$; for example, $A = H^{-1/2} B^{1/2}$. Define, near ξ_1 ,

$$\begin{aligned} \psi(\xi) &= h_0(A(\xi - \xi_1)) + q(\xi_1) - h_0(0) \\ &\quad + (\nabla q(\xi_1) - A^\top \nabla h_0(0)) \cdot (\xi - \xi_1). \end{aligned}$$

Thus ψ and q have the same value, gradient, and Hessian at ξ_1 . This change of graph is induced by an invertible affine map of ambient frequency space: its base part is $x \mapsto \xi_1 + A^{-1}x$, and its height part is addition of an affine function of the new base coordinates.

Let $\vartheta \in C_c^\infty(B(0, 2))$ equal one on $B(0, 1)$ and take $0 \leq \vartheta \leq 1$. For a small fixed $\epsilon > 0$ put

$$R_\epsilon(\xi) = \vartheta((\xi - \xi_1)/\epsilon)(\psi(\xi) - q(\xi)).$$

Choose ϵ so that its support is contained in V and the target graph is defined there. Taylor's theorem and the agreement of the two-jets give

$$\|D^j R_\epsilon\|_\infty \leq C\epsilon^{3-j}, \quad j = 0, 1, 2.$$

Consequently $\phi = q + R_\epsilon$ remains uniformly convex for sufficiently small ϵ and agrees with ψ on $B(\xi_1, \epsilon)$. It equals Q_s outside a compact set, so all of its derivatives of order at least two are bounded.

To verify the required continuation form, observe that $D = q_s - Q_s$ is strictly positive for $0 < |\xi| < 1$. Shrink V , if necessary, so that $D \geq d_0 > 0$ there. The function R_ϵ/D , extended by zero outside V , is smooth. Set

$$\tilde{\zeta} = \zeta_0 + R_\epsilon/D.$$

On the perturbation support, $\zeta_0 = 1/2$ and $|R_\epsilon/D| \leq C\epsilon^3/d_0$. Choosing ϵ smaller makes this last quantity at most $1/4$. Hence $0 \leq \tilde{\zeta} \leq 1$, with the same inner and outer support properties as ζ_0 , and $\phi = Q_s + \tilde{\zeta}(q_s - Q_s)$. Every choice was made using only the fixed target graph. $\square$

In what follows, the phases to which the cited packet theorem is applied are chosen as in Lemma 3.1. Although the inserted graph lies in the transition annulus, the packet theorem allows arbitrary frequency labels in a fixed bounded region. In particular, those labels need not lie in the spherical region $B(0, \rho)$: [11, Definition 6.3] imposes no such restriction, and its single-step maps act on arbitrary finite input arrays. All windows meeting the inserted graph, and the terminal windows used below, can therefore be included in one fixed bounded frequency region. The constants may depend on the chosen phase, including its higher derivative bounds.

3.2 Frames and the capacity norm

We use the standard frequency and spatial localization of wave packets; see Tao [17, arXiv version 2, Section 4] and the Fourier-series construction of Guo, Oh, Wang, Wu, and Zhang [6, arXiv version 1, Section 4.1]. Square windows give the exact identities below, with the normalization needed for our sample sum.

Fix a dyadic $N \geq 2$ and put $\delta = N^{-1}$ and $U_\nu = -\nabla\phi(\nu)$. For a dyadic factor $1 \leq m \leq N$ let

$$\theta = m\delta, \quad r = m^{-2}, \quad w = r\theta = \delta/m. \quad (3.2)$$

Partition $[0, 1)$ into intervals I of length r , with left endpoints t_I . At this scale, the indices are

$$(\nu, z, I), \quad \nu \in \theta\mathbb{Z}^2, \quad z \in D_f^{-1}w\mathbb{Z}^2.$$

Using the windows fixed in the local setup, define

$$(\mathbf{P}_{m,I}f)_{\nu,z} = \int f(\xi) \chi_\theta^\nu(\xi) e(N^2(z \cdot \xi + t_I\phi(\xi))) d\xi, \quad (3.3)$$

$$S_{m,I}d(\xi) = (D_f\theta)^{-2} \sum_{\nu,z} d_{\nu,z} \chi_\theta^\nu(\xi) e(-N^2(z \cdot \xi + t_I\phi(\xi))). \quad (3.4)$$

Synthesis is initially defined for finite arrays and then extended continuously to ℓ^2 .

Lemma 3.2 (Exact frame identities). *For every $f \in L^2(\mathbb{R}^2)$,*

$$S_{m,I}\mathbf{P}_{m,I}f = f, \quad \|\mathbf{P}_{m,I}f\|_{\ell^2}^2 = (D_f\theta)^2 \|f\|_2^2, \quad \|\mathbf{S}_{m,I}\|_{\ell^2 \rightarrow L^2} = (D_f\theta)^{-1}. \quad (3.5)$$

Proof. Since $N^2 w \theta = 1$, the vectors $N^2 z$ run through $(D_f \theta)^{-1} \mathbb{Z}^2$. For each fixed ν , Equation (3.3) is the Fourier coefficient array of $f \chi_\theta^\nu e(N^2 t_I \phi)$ on a square of side $D_f \theta$. Parseval and Fourier inversion give

$$\sum_z |(\mathbf{P}_{m,I} f)_{\nu,z}|^2 = (D_f \theta)^2 \|f \chi_\theta^\nu\|_2^2$$

and, after synthesis, $f(\chi_\theta^\nu)^2$. Sum over ν and use $\sum_\nu (\chi_\theta^\nu)^2 = 1$. The norm of synthesis follows from $\mathbf{S}_{m,I} = (D_f \theta)^{-2} \mathbf{P}_{m,I}^*$. These identities first hold for finite partial Fourier sums in L^2 and then for their limits. $\square$

For $0 < \kappa < 1/10$, let E be a centered closed ellipse with ordered semiaxes $L \geq a > 0$, of any orientation, and write $W_\kappa(E) = L^{1+\kappa} a^{1-\kappa}$. For a finite array c in one bin at the scale (θ, r, w) define

$$K_{\theta,r}(c) = \sup_{\substack{E,u,v \\ L \geq a \geq \theta}} \frac{\theta^2}{W_\kappa(E)} \sum_{\substack{U_\nu \in u+E \\ z \in v+rE}} |c_{\nu,z}|^2, \quad (3.6)$$

$$G_{\theta,r}(c)^3 = \left(w^2 \sum_{\nu,z} |c_{\nu,z}|^2 \right) K_{\theta,r}(c)^{1/2}. \quad (3.7)$$

The translations $u, v \in \mathbb{R}^2$ are independent, and there is no upper restriction on either semiaxis. On each finite index set let

$$A_{\theta,r}(c) = \inf \left\{ \sum_{\ell=1}^q G_{\theta,r}(a_\ell) : |c| \leq \sum_{\ell=1}^q |a_\ell|, \ q < \infty \right\},$$

$$\mathcal{A}_m(c)^3 = r \sum_I A_{\theta,r}(c_I)^3.$$

The domination in this infimum is coordinatewise.

Lemma 3.3 (The atomic norm controls the sample sum). *The function $A_{\theta,r}$ is a norm on every finite coordinate space, is monotone under absolute coordinatewise domination, and is contracted by coordinate masks. Moreover,*

$$r w^2 \sum_{\nu,z,I} |c_{\nu,z,I}|^3 \leq \mathcal{A}_m(c)^3. \quad (3.8)$$

Proof. The disk of radius θ , with its two translations chosen independently, contains any prescribed index in both tests. Its weight is θ^2 , so $K_{\theta,r}(c) \geq \|c\|_\infty^2$ and

$$G_{\theta,r}(c)^3 \geq w^2 \|c\|_2^2 \|c\|_\infty \geq w^2 \sum_{\nu,z} |c_{\nu,z}|^3.$$

For every atomic domination of c , the triangle inequality in ℓ^3 now gives

$$w^{2/3} \|c\|_3 \leq \sum_\ell G_{\theta,r}(a_\ell).$$

Taking the infimum and summing its cube over bins with weight r proves Equation (3.8). A singleton has cost $w^{2/3} |c_{\nu,z}|$, so the infimum is finite; the displayed lower bound makes it positive away from zero. Homogeneity, subadditivity, and absolute monotonicity follow directly from the definition, as does contraction under masks. $\square$

At the initial scale $m = 1$ there is one bin, $[0, 1)$, and we use the notation

$$\mathcal{K}_\kappa(d) = K_{\delta,1}(d), \quad m(d) = \delta^2 \sum_{\nu,z} |d_{\nu,z}|^2. \quad (3.9)$$

In particular, $G_{\delta,1}(d)^3 = m(d) \mathcal{K}_\kappa(d)^{1/2}$.

3.3 The cited theorem and its specialization

A single-step map T_m synthesizes at factor 1, analyzes in every factor- m time bin, and then applies any coordinate mask. It uses the full matrix between its input and retained output indices. The admissible complexity convention of [11, Definition 6.3] fixes a finite B and a bounded frequency-label region, requires $m \leq N \leq m^B$ and position labels of size at most m^B , and allows arbitrary finite arrays on those indices. The phase and window constants are fixed throughout this class.

Theorem 3.4 (Packet propagation [11, Theorem 6.4, Equation (6.8)]). *For a fixed phase as in Equation (3.1), every fixed admissible complexity range, $0 < \kappa < 1/10$, and $\eta > 0$, there is a finite constant C such that*

$$\mathcal{A}_m(T_m d)^3 \leq C m^{10\kappa+\eta} G_{\delta,1}(d)^3 \quad (3.10)$$

for all its single-step maps and all its finite complex input arrays. The source also permits $A_{\delta,1}(d)^3$ on the right. The constant may depend on κ , η , the complexity range, the phase, and the windows.

In particular, suppose the initial labels $\nu \in \delta\mathbb{Z}^2$ and $z \in D_f^{-1}\delta\mathbb{Z}^2$ lie in fixed bounded sets. Define

$$Sd(\xi) = (D_f\delta)^{-2} \sum_{\nu,z} d_{\nu,z} \chi_\delta^\nu(\xi) e(-N^2 z \cdot \xi), \quad (3.11)$$

$$c_{\nu',z',j} = \int Sd(\xi) \chi_1^{\nu'}(\xi) e(N^2 z' \cdot \xi + j\phi(\xi)) d\xi, \quad (3.12)$$

where $\nu' \in \mathbb{Z}^2$, $z' \in D_f^{-1}N^{-2}\mathbb{Z}^2$, and $j = 0, \dots, N^2 - 1$. Retain any output coordinate mask for which ν', z' lie in fixed bounded sets. There is a finite $P(\kappa, \eta) \geq 1$, depending on these fixed sets, the phase, and the windows, such that, for every dyadic $N \geq 2$ and $0 < \eta \leq 1$,

$$N^{-6} \sum_{\nu',z',j} |c_{\nu',z',j}|^3 \leq P(\kappa, \eta) N^{10\kappa+\eta} m(d) \mathcal{K}_\kappa(d)^{1/2}. \quad (3.13)$$

The sum is over the retained coordinates.

Specialization of the cited theorem. Set $m = N$. The scales in Equation (3.2) become

	θ	r	w	$t_I = \frac{j}{N^2}, \quad rw^2 = N^{-6} \quad \text{at output.}$
input	δ	1	δ	
output	1	N^{-2}	N^{-2}	

Thus the companion's synthesis and analysis operators are exactly the displayed operators, with its window constant L_f equal to our D_f . The initial gauge is exactly the last expression in Equation (3.13), and Equation (3.8) gives

$$N^{-6} \sum_{\nu',z',j} |c_{\nu',z',j}|^3 \leq \mathcal{A}_N(c)^3.$$

All bounded label sets fit one fixed admissible complexity range for sufficiently large N , since $m = N$; the finitely many smaller dyadic scales may be included by enlarging the constant. The bound at any such fixed scale follows from finite-dimensional boundedness and $\mathcal{K}_\kappa(d) \geq \|d\|_\infty^2$. Applying Equation (3.10) proves Equation (3.13). This argument uses arbitrary input arrays, so it also applies after any restriction of their support. $\square$

Theorem 3.4 is the additional oscillatory input. Its source proves Equation (3.10) in [11, Section 7]; no rate for the dependence of $P(\kappa, \eta)$ on its small parameters is asserted here.

4 From packet capacity to weighted tubes

We first obtain a positive tube estimate from theakeya maximal theorem. We then use a decomposition by capacity to apply Theorem 3.4 to an arbitrary initial array without imposing a condition on the sizes of its coefficients. Throughout this section, the frequency and position label bounds are fixed as in the packet setup. The positive tube-duality formulation follows the classical mixed-norm framework of Tao, Vargas, and Vega [18, arXiv version 1, Section 3, Equation (21)]; the capacity decomposition below is proved here.

For an initial array $d = (d_{\nu,z})$, define

$$\begin{aligned} T_{\nu,z} &= \{(X, t) \in \mathbb{R}^2 \times \mathbb{R} : 0 \leq t \leq 1, |X - z - tU_\nu| \leq \delta\}, \\ H_d(X, t) &= \sum_{\nu,z} |d_{\nu,z}|^2 \mathbf{1}_{T_{\nu,z}}(X, t), \quad b_\nu(d) = \sum_z |d_{\nu,z}|^2. \end{aligned} \quad (4.1)$$

All integrals of tube functions use Lebesgue measure $dX dt$. In particular, $|T_{\nu,z}| = \pi\delta^2$ and

$$\int_{\mathbb{R}^3} H_d = \pi m(d). \quad (4.2)$$

For clarity, the precise maximal operator used below is

$$K_\rho g(\omega) = \sup_{a \in \mathbb{R}^3} \frac{1}{\pi\rho^2} \int_{\mathcal{T}_\rho(a, \omega)} |g(y)| dy, \quad \mathcal{T}_\rho(a, \omega) = \{a + r\omega + w : |r| \leq \frac{1}{2}, w \perp \omega, |w| \leq \rho\}.$$

Thus $\mathcal{T}_\rho(a, \omega)$ is a circular cylinder of length one and volume $\pi\rho^2$. The companion theorem [12, Theorem 1.1] states that for every $s > 0$ there is a finite constant $M(s)$, which we enlarge to be at least one, such that for every $g \in L^3(\mathbb{R}^3)$,

$$\|K_\rho g\|_{L^3(S^2)} \leq M(s)\rho^{-s}\|g\|_{L^3(\mathbb{R}^3)}, \quad 0 < \rho < 1. \quad (4.3)$$

Here S^2 carries its usual surface measure. We will use only $0 < s \leq 1$.

Lemma 4.1 (Weightedakeya estimate). *There are fixed constants C and N_0 such that for every dyadic $N \geq N_0$, every initial array d , and every $0 < s \leq 1$,*

$$\int H_d^{3/2} \leq CM(s)^{3/2} \delta^{-3s/2} \delta^2 \sum_\nu b_\nu(d)^{3/2}. \quad (4.4)$$

The constants C, N_0 depend only on the fixed geometric setup.

Proof. By Equation (2.4), the velocity map is bi-Lipschitz on the fixed frequency-label range, where it is bounded. The map $U \mapsto (U, 1)/\sqrt{1 + |U|^2}$ is bi-Lipschitz on this bounded range: its inverse is $\omega \mapsto \omega'/\omega_3$, whose derivative is bounded where ω_3 is bounded away from zero. Thus the directions

$$\omega_\nu = \frac{(U_\nu, 1)}{\sqrt{1 + |U_\nu|^2}}$$

are separated by a fixed positive multiple of δ . Choose spherical caps Ω_ν centered at ω_ν with radius $c_1\delta$, where the fixed $c_1 > 0$ is small enough that these caps are disjoint. Their surface areas satisfy $c_2\delta^2 \leq |\Omega_\nu| \leq C_2\delta^2$.

The center segment of $T_{\nu,z}$ has Euclidean length $\sqrt{1 + |U_\nu|^2}$, bounded by a fixed constant. For each $\omega \in \Omega_\nu$, its distance from the line through $(z, 0)$ in direction ω is $O(\delta)$. Adding the horizontal

cross-section of radius δ preserves this bound. The projection of the entire tube onto that line is an interval of bounded length. Cover that interval by a bounded number of intervals of length one. It follows that $T_{\nu,z}$ is contained in at most C_3 cylinders $\mathcal{T}_{C_4\delta}(a, \omega)$, where C_3 and $C_4 \geq 1$ are fixed. In particular, for every $g \in L^3(\mathbb{R}^3)$,

$$\int_{T_{\nu,z}} |g| \leq C\delta^2 K_{C_4\delta} g(\omega) \quad (\omega \in \Omega_\nu).$$

Averaging over Ω_ν , multiplying by $|d_{\nu,z}|^2$, and summing gives

$$\int H_d |g| \leq C \sum_\nu b_\nu(d) \int_{\Omega_\nu} K_{C_4\delta} g(\omega) d\omega.$$

Set $B_d = \sum_\nu b_\nu(d) \mathbf{1}_{\Omega_\nu}$. Disjointness of the caps gives

$$\|B_d\|_{L^{3/2}(S^2)} \leq C \left(\delta^2 \sum_\nu b_\nu(d)^{3/2} \right)^{2/3}.$$

Choose N_0 so that $C_4\delta < 1$. Hölder's inequality on S^2 and Equation (4.3) now yield

$$\int H_d |g| \leq CM(s) \delta^{-s} \left(\delta^2 \sum_\nu b_\nu(d)^{3/2} \right)^{2/3} \|g\|_3.$$

We used $(C_4\delta)^{-s} \leq \delta^{-s}$, so the displayed constant is independent of s . Duality between $L^{3/2}$ and L^3 , followed by taking the power $3/2$, proves Equation (4.4). The sum over positions causes no loss because K_ρ already takes the supremum over cylinder locations. $\square$

The next observation compares a single ellipse test with the tube mass. Recall that E is a centered ellipse with semiaxes $L \geq a \geq \delta$, and

$$W_\kappa(E) = L^{1+\kappa} a^{1-\kappa} = La(L/a)^\kappa \geq La.$$

If a nonzero coordinate subarray q of d is supported on $U_\nu \in u + E$, $z \in v + E$, then for $0 \leq t \leq 1$ the horizontal support of $H_q(\cdot, t)$ lies in

$$v + tu + E + tE + B(0, \delta) \subset v + tu + 3E.$$

Indeed, $B(0, \delta) \subset E$, and the convexity and central symmetry of E imply $E + tE = (1+t)E$. Therefore $|\text{supp } H_q| \leq 9\pi La$. Equation (4.2) and Hölder's inequality give

$$\int H_q^{3/2} \geq \frac{(\int H_q)^{3/2}}{|\text{supp } H_q|^{1/2}} \geq \frac{\pi}{3} m(q) \left(\frac{m(q)}{W_\kappa(E)} \right)^{1/2}. \quad (4.5)$$

The bound is uniform in the eccentricity L/a , in the independent translations u, v , and in $\kappa > 0$.

Lemma 4.2 (Decomposition by capacity). *Let d be a nonzero finite initial array, and let $n = \#\text{supp } d$. For every $0 < \kappa < 1/10$, there is a decomposition into coordinate subarrays with pairwise disjoint supports,*

$$d = \sum_{i=1}^J d^{(i)}, \quad J \leq C \log(2+n),$$

such that

$$\sum_{i=1}^J m(d^{(i)}) \mathcal{K}_\kappa(d^{(i)})^{1/2} \leq C \int H_d^{3/2}. \quad (4.6)$$

Both constants are absolute and independent of the nonzero coefficient sizes and of κ .

Proof. Write $D = \max_{\nu, z} |d_{\nu, z}|^2 > 0$. A disk of radius δ , independently translated to a prescribed velocity and position, is an admissible test with weight δ^2 . On the other hand, every test has $W_\kappa(E) \geq \delta^2$. Hence

$$D \leq \mathcal{K}_\kappa(d) \leq nD. \quad (4.7)$$

Capacity is monotone under restriction to a subset of coordinates.

Put $A = D/n^2$. The cost of any remaining array with capacity at most A is controlled by the contribution of a largest original coefficient to the tube integral. Indeed, any coordinate subarray r with $\mathcal{K}_\kappa(r) \leq A$ satisfies

$$m(r)\mathcal{K}_\kappa(r)^{1/2} \leq \delta^2 n D \sqrt{D/n^2} = \delta^2 D^{3/2}.$$

An original entry whose squared modulus is D contributes $D\mathbf{1}_{T_{\nu, z}}$ to H_d , whence $\int H_d^{3/2} \geq \pi\delta^2 D^{3/2}$. Thus the cost of any such remainder is controlled by $\pi^{-1} \int H_d^{3/2}$, independently of which entries remain.

Start with remainder $r_1 = d$. Whenever $\mathcal{K}_\kappa(r_j) > A$, choose an ellipse test whose density is at least half of that supremum. Let q_j contain all nonzero coordinates of r_j selected by that test, write E_j for its ellipse, and put

$$\rho_j = \frac{m(q_j)}{W_\kappa(E_j)} \geq \frac{1}{2}\mathcal{K}_\kappa(r_j), \quad r_{j+1} = r_j - q_j.$$

Such a test exists without assuming that the supremum is attained. It selects at least one nonzero entry, so the process terminates after at most n removals. Denote the terminal remainder by r_* ; then $\mathcal{K}_\kappa(r_*) \leq A$, including the possibility $r_* = 0$.

We group the removed bundles by the capacity of the remainder from which they were selected. This controls the capacity of each group by its first remainder, while leaving only logarithmically many groups. For each integer $k \geq 0$, set $\lambda_k = 2^k A$, and group together all q_j for which

$$\lambda_k \leq \mathcal{K}_\kappa(r_j) < 2\lambda_k.$$

Let d_k be the union array at this level, omitting empty levels. The array d_k is a coordinate restriction of the remainder at the first removal in that level. Therefore $\mathcal{K}_\kappa(d_k) < 2\lambda_k$. Every bundle at this level has $\rho_j \geq \lambda_k/2$, so additivity of m gives

$$\begin{aligned} m(d_k)\mathcal{K}_\kappa(d_k)^{1/2} &\leq \sqrt{2\lambda_k} \sum_{j \text{ at level } k} m(q_j) \\ &\leq 2 \sum_{j \text{ at level } k} m(q_j) \sqrt{\rho_j}. \end{aligned}$$

Use Equation (4.5) for each selected test. Since the q_j are disjoint coordinate subarrays, their tube functions sum to a function bounded by H_d . For nonnegative numbers, $\sum_j x_j^{3/2} \leq (\sum_j x_j)^{3/2}$. Consequently

$$\sum_k m(d_k)\mathcal{K}_\kappa(d_k)^{1/2} \leq C \sum_j \int H_{q_j}^{3/2} \leq C \int H_d^{3/2}. \quad (4.8)$$

The stopping estimate applies to r_* . Adding its cost to Equation (4.8) proves Equation (4.6).

Finally, at every removal the capacity lies in $(A, nD]$, by monotonicity and Equation (4.7). The ratio of these endpoints is n^3 , so there are at most $1 + \lfloor 3 \log_2 n \rfloor$ occupied dyadic levels. Including the nonzero remainder gives $J \leq C \log(2 + n)$. If $n = 1$, then $\mathcal{K}_\kappa(d) = D = A$; the process makes no removal and the remainder alone gives the conclusion. No lower bound on the smallest coefficient has been used. $\square$

Proposition 4.3 (Discrete critical estimate). *Let c be the terminal coefficient array obtained from d by the synthesis and sampling map of Theorem 3.4, with any fixed permitted output mask. For every sufficiently large dyadic N , $0 < \kappa < 1/10$, $0 < \eta \leq 1$, and $0 < s \leq 1$, put $\gamma_0 = 10\kappa + \eta + 3s/2$. Then*

$$\sum_{\nu', z', j} |c_{\nu', z', j}|^3 \leq CN^6 P(\kappa, \eta) M(s)^{3/2} N^{\gamma_0} \log^2(2 + N) \delta^2 \sum_{\nu} b_{\nu}(d)^{3/2}. \quad (4.9)$$

The constant C and the lower threshold for N are independent of κ, η, s . Their dependence is only on the fixed phase, windows, and label bounds; all additional parameter dependence is contained in $P(\kappa, \eta)$ and $M(s)$.

Proof. There is nothing to prove if $d = 0$. Otherwise use the decomposition in Lemma 4.2, and let $c^{(i)}$ be the terminal array of $d^{(i)}$ with the same output mask. The sampling map is linear, so $c = \sum_i c^{(i)}$. The triangle inequality in ℓ^3 , Hölder's inequality for the finite sum over i , and Theorem 3.4 give

$$\begin{aligned} N^{-6} \sum |c|^3 &\leq J^2 \sum_i N^{-6} \sum |c^{(i)}|^3 \\ &\leq J^2 P(\kappa, \eta) N^{10\kappa+\eta} \sum_i m(d^{(i)}) \mathcal{K}_{\kappa}(d^{(i)})^{1/2} \\ &\leq C \log^2(2 + n) P(\kappa, \eta) N^{10\kappa+\eta} \int H_d^{3/2}. \end{aligned}$$

There are $O(N^2)$ initial frequency labels and $O(N^2)$ initial position labels in the fixed bounded ranges, so $n \leq CN^4$. It follows that

$$N^{-6} \sum |c|^3 \leq CP(\kappa, \eta) N^{10\kappa+\eta} \log^2(2 + N) \int H_d^{3/2}. \quad (4.10)$$

Insert Lemma 4.1 and use $\delta^{-3s/2} = N^{3s/2}$ to obtain Equation (4.9). Every coordinate restriction used above obeys the same label bounds as d , so the same packet constant $P(\kappa, \eta)$ applies to every piece. $\square$

5 Joint estimates on sparse balls

The factor N^{γ_0} in Proposition 4.3 is a small loss on a bounded region. To remove it later, we need an estimate for many distant regions with a single data norm on the right. The directions in which two translations fail to oscillate are confined to a small frequency set. Discarding that set costs little in L^∞ ; the remaining translations are almost orthogonal at the initial packet scale. This joint estimate supplies the input to the sparse-ball strategy for removing a spatial loss [3, 8, 16]; we prove the required estimate for L^3 data with the precise separation and uniformity below.

Proposition 5.1 (Sparse balls). *There is a constant $\mu_0 \in (0, 1)$ depending only on the fixed graph setup with the following property. Suppose that h is supported in Q_0 , $\|h\|_3 \leq 1$, and $0 < \mu \leq \mu_0$. Let $\mathcal{B}$ be a finite family of balls of common radius R such that*

$$\#\mathcal{B} \leq \mu^{-8}, \quad R \geq \mu^{-100}, \quad |x_B - x_{\tilde{B}}| \geq \mu^{-40} R \quad (B \neq \tilde{B}),$$

where x_B denotes the center of B . Choose a dyadic integer N with $4R \leq N^2 < 16R$. For $0 < \kappa < 1/10$ and $0 < \eta, s \leq 1$, put $\gamma_0 = 10\kappa + \eta + 3s/2$. Then

$$\left| \{|F| > \mu\} \cap \bigcup_{B \in \mathcal{B}} B \right| \leq C \mu^{-3} P(\kappa, \eta) M(s)^{3/2} N^{\gamma_0} \log^2(2 + N). \quad (5.1)$$

The constant C and the threshold μ_0 are independent of κ, η, s , of the centers, and of their maximum separation.

Proof. Write $\delta = N^{-1}$ and $D = D_f$. Every window χ_δ^ν meeting Q_0 lies in a fixed compact neighborhood Q_1 of Q_0 , since $\delta \leq 1$. Fix

$$G = \sup_{Q_1} |\nabla \phi|, \quad Z > 4(2 + G).$$

The initial spatial labels will satisfy $|z| \leq Z$ and the terminal spatial labels $|z'| \leq 1$. Thus all label bounds used in Proposition 4.3 are fixed independently of the small exponents. Throughout the proof we decrease μ_0 , if necessary, only in terms of this fixed setup. In particular, we may require N to exceed any fixed lower threshold needed below. The hypotheses give

$$\delta \leq \frac{1}{2}\mu^{50}, \quad \mu^{-8} \leq N, \quad H \geq \mu^{-40}R > \frac{1}{16}\mu^{-40}N^2 \quad (5.2)$$

whenever H is the separation of two distinct centers.

Removing resonant frequencies. Set $a_B = x_B - (0, 0, N^2/2)$. For each pair of distinct balls write $a_B - a_{\tilde{B}} = (q, s_0) \in \mathbb{R}^2 \times \mathbb{R}$ and $H = |(q, s_0)|$. Its resonant set is

$$\Omega_{B, \tilde{B}} = \{\xi \in Q_0 : |q + s_0 \nabla \phi(\xi)| \leq H\mu^{20}\}.$$

If this set is nonempty, then

$$H \leq |q| + |s_0| \leq H\mu^{20} + (G + 1)|s_0|,$$

so $|s_0| \geq H/[2(G + 1)]$ for sufficiently small μ_0 . For two points of $\Omega_{B, \tilde{B}}$, Equation (2.4) gives

$$c|\xi - \zeta| \leq 2H\mu^{20}/|s_0| \leq C\mu^{20}.$$

Consequently each resonant set has area at most $C\mu^{40}$. The union Ω over all pairs therefore has area at most $C\mu^{24}$. Define $h_* = h\mathbf{1}_{Q_0 \setminus \Omega}$, and let F_* be the extension defining F with h replaced by h_* . By Hölder's inequality,

$$\|F - F_*\|_\infty \leq \|\chi_1^{\nu_*}\|_\infty \|h\|_3 |\Omega|^{2/3} \leq C\mu^{16} \leq \mu/2. \quad (5.3)$$

The powers 20, 40, and 100 provide ample room both for this discarded mass and for the phase estimates below; their precise values are not essential.

A joint bound for the initial coefficients. Let

$$\mathcal{U} = [0, D^{-1})^2 \times [0, 1), \quad \Gamma_\nu = \text{supp } \chi_\delta^\nu.$$

For $u \in \mathcal{U}$ and $B \in \mathcal{B}$ define

$$\begin{aligned} h_{B,u}(\xi) &= h_*(\xi) e((a_B + u) \cdot (\xi, \phi(\xi))), \\ d_{\nu,z}^{B,u} &= \int h_{B,u}(\xi) \chi_\delta^\nu(\xi) e(N^2 z \cdot \xi) d\xi, \quad z \in D^{-1}\delta\mathbb{Z}^2, \\ b_\nu^{B,u} &= \sum_{|z| \leq Z} |d_{\nu,z}^{B,u}|^2. \end{aligned} \quad (5.4)$$

Only angular windows meeting Q_0 need be retained. We claim that, uniformly in u ,

$$\sum_{B \in \mathcal{B}} b_\nu^{B,u} \leq C\delta^2 \int_{\Gamma_\nu} |h_*|^2. \quad (5.5)$$

If $h_*\chi_\delta^\nu = 0$ almost everywhere, the claim is immediate. Otherwise choose a point $\xi_g \in \Gamma_\nu \cap (Q_0 \setminus \Omega)$. For this ν , define the operator on *unrestricted* $L^2(\mathbb{R}^2)$ by

$$(T_\nu f)_{B,z} = \int f(\xi) \chi_\delta^\nu(\xi) e((a_B + u) \cdot (\xi, \phi(\xi)) + N^2 z \cdot \xi) d\xi, \quad |z| \leq Z.$$

We will apply it to $f = h_*\mathbf{1}_{\Gamma_\nu}$. This enlargement of the operator domain is useful: the kernel of $T_\nu T_\nu^*$ has the smooth amplitude $(\chi_\delta^\nu)^2$, with no indicator of the excised set.

For a single ball block, Parseval's identity on the square of side $D\delta$ containing Γ_ν gives

$$\sum_{z \in D^{-1}\delta\mathbb{Z}^2} |(T_\nu f)_{B,z}|^2 = (D\delta)^2 \int |f(\xi) \chi_\delta^\nu(\xi)|^2 d\xi \leq C\delta^2 \|f\|_2^2.$$

Here the full spatial sum is understood before its restriction to $|z| \leq Z$; its Fourier frequencies are $N^2 z \in (D\delta)^{-1}\mathbb{Z}^2$. It follows that each diagonal ball block of $T_\nu T_\nu^*$ has operator norm at most $C\delta^2$.

For distinct $B, \tilde{B}$, an entry of this Gram operator is

$$\int (\chi_\delta^\nu(\xi))^2 e((q + N^2(z - \tilde{z})) \cdot \xi + s_0 \phi(\xi)) d\xi.$$

To estimate it, put $\xi = \nu + \delta v$ and denote the resulting phase by $\Psi(v)$. Let

$$v_0 = \frac{q + s_0 \nabla \phi(\xi_g)}{|q + s_0 \nabla \phi(\xi_g)|}, \quad \Lambda = \delta H \mu^{20}.$$

The direction v_0 is defined because ξ_g avoids every resonant set. On the entire support of the smooth amplitude $\chi(v)^2$,

$$D_{v_0} \Psi(v) \geq \delta(H\mu^{20} - CH\delta - 2ZN^2) \geq \Lambda/2, \quad (5.6)$$

$$|\partial^\alpha \Psi(v)| \leq C_\alpha H \delta^2 \leq C_\alpha \Lambda \quad (|\alpha| \geq 2), \quad \Lambda \geq \frac{1}{16} N \mu^{-20}. \quad (5.7)$$

For Equation (5.6), the two errors relative to $H\mu^{20}$ are at most $C\mu^{30}$ and $32Z\mu^{20}$ by Equation (5.2); their sum is at most $1/2$ after fixing μ_0 . Equation (5.7) follows from the bounded higher derivatives of ϕ and $\delta\mu^{-20} \leq \mu^{30}/2$.

Repeated integration by parts in the fixed direction v_0 now gives, for each fixed integer $k \geq 1$,

$$\left| \int (\chi_\delta^\nu)^2 e((q + N^2(z - \tilde{z})) \cdot \xi + s_0 \phi(\xi)) d\xi \right| \leq C_k \delta^2 \Lambda^{-k} \leq C_k \delta^2 N^{-k}. \quad (5.8)$$

For completeness, use the operator $(2\pi i D_{v_0} \Psi)^{-1} D_{v_0}$ on the exponential. The lower bound for $D_{v_0} \Psi$ and the higher derivative bounds imply that each fixed directional derivative of $(D_{v_0} \Psi)^{-1}$ is $O_k(\Lambda^{-1})$. Applying the adjoint k times to the fixed smooth bump therefore costs $O_k(\Lambda^{-k})$. No upper bound on the first derivative is required. In particular, these constants remain uniform as H becomes arbitrarily large: the lower bound and the higher derivative bounds carry the same factor H .

There are at most $C_Z N^2$ retained spatial labels in each ball block. Thus every row and column of the off-diagonal block matrix has at most $C_Z \mu^{-8} N^2 \leq C_Z N^3$ entries. Taking $k = 4$ in Equation (5.8) and applying the row and column sum bound for the ℓ^2 operator norm shows that its norm is at most $C\delta^2$. Together with the diagonal block estimate, this gives $\|T_\nu\|_{2 \rightarrow 2}^2 \leq C\delta^2$. Applying it to $h_*\mathbf{1}_{\Gamma_\nu}$ proves Equation (5.5).

Since the windows have bounded overlap and $|\Gamma_\nu| \leq C\delta^2$, Hölder's inequality gives the precise power needed for Proposition 4.3:

$$\begin{aligned} \delta^2 \sum_{\nu, B} (b_\nu^{B,u})^{3/2} &\leq \delta^2 \sum_\nu \left(\sum_B b_\nu^{B,u} \right)^{3/2} \leq C\delta^5 \sum_\nu \left(\int_{\Gamma_\nu} |h_*|^2 \right)^{3/2} \\ &\leq C\delta^5 \sum_\nu |\Gamma_\nu|^{1/2} \int_{\Gamma_\nu} |h_*|^3 \leq C\delta^6. \end{aligned} \quad (5.9)$$

Reconstruction and the discarded spatial tail. Let $d^{B,u}$ now denote the finite array in Equation (5.4) restricted to $|z| \leq Z$. Apply the synthesis and sampling map to this array, retaining only $\nu' = \nu_*$, $|z'| \leq 1$, and $0 \leq j < N^2$. Write the resulting array as $c_{z',j}^{B,u}$. We next justify, uniformly in B, u, z', j ,

$$c_{z',j}^{B,u} = F_*(a_B + u + (N^2 z', j)) + O(N^{-3}). \quad (5.10)$$

To do so, write $\widehat{d}^{B,u}$ for the initial analysis array with the spatial grid untruncated. Lemma 3.2 at the initial scale gives

$$(D\delta)^{-2} \sum_{\nu, z} \widehat{d}_{\nu, z}^{B,u} \chi_\delta^\nu(\xi) e(-N^2 z \cdot \xi) = h_{B,u}(\xi) \quad \text{in } L^2(\mathbb{R}^2). \quad (5.11)$$

The same lemma gives the exact squared-mass identity

$$\sum_{\nu, z} |\widehat{d}_{\nu, z}^{B,u}|^2 = (D\delta)^2 \|h_{B,u}\|_2^2 = (D\delta)^2 \|h_*\|_2^2 \leq C\delta^2. \quad (5.12)$$

These statements require only $h \in L^2(Q_0)$, which follows from the assumed L^3 bound and compact support. In particular, the possibly very large center modulation does not affect the bound.

A matrix entry from an omitted $|z| > Z$ to a retained output is

$$A_{z',j;\nu,z} = (D\delta)^{-2} \int \chi_\delta^\nu(\xi) \chi_1^{\nu*}(\xi) e(N^2(z' - z) \cdot \xi + j\phi(\xi)) d\xi.$$

After the substitution $\xi = \nu + \delta v$, the prefactor is D^{-2} , the amplitude has uniformly bounded smooth derivatives, and the derivative of the phase in the direction $-z/|z|$ is at least

$$N(|z| - 1 - G) \geq cN(1 + |z|).$$

All phase derivatives of order at least two are bounded uniformly, because $j\delta^2 \leq 1$. The same integration by parts argument therefore gives

$$|A_{z',j;\nu,z}| \leq C_k [N(1 + |z|)]^{-k}. \quad (5.13)$$

There are $O(N^2)$ angular labels, and on the spatial mesh $D^{-1}N^{-1}\mathbb{Z}^2$,

$$\sum_{|z| > Z} (1 + |z|)^{-2k} \leq C_k N^2 \quad (k > 2).$$

Consequently each discarded part of a matrix row has squared ℓ^2 norm at most $C_k N^{4-2k}$. By Equation (5.12), its pairing with $\widehat{d}^{B,u}$ is at most $C_k N^{1-k}$. Taking $k = 4$ proves Equation (5.10). One may justify all infinite sums by the L^2 convergence in Equation (5.11); the tail pairing is also absolutely convergent by Cauchy–Schwarz. Thus neither the data nor the sharp excision boundary has been differentiated.

Summing samples and filling the balls. Apply Proposition 4.3 to each retained array and sum over B . Equation (5.9) cancels its factor N^6 exactly and gives

$$\sum_{B, z', j} |c_{z', j}^{B, u}|^3 \leq CP(\kappa, \eta) M(s)^{3/2} N^{\gamma_0} \log^2(2 + N).$$

There are $O(N^4)$ terminal spatial labels and N^2 times per ball, so the total number of samples is $O(\mu^{-8} N^6) = O(N^7)$. The cubed errors from Equation (5.10) sum to at most $CN^7 N^{-9} = CN^{-2}$. Using $|a + b|^3 \leq 4(|a|^3 + |b|^3)$ and $P, M \geq 1$, we obtain

$$\sum_{B, z', j} |F_*(a_B + u + (N^2 z', j))|^3 \leq CP(\kappa, \eta) M(s)^{3/2} N^{\gamma_0} \log^2(2 + N), \quad (5.14)$$

uniformly for $u \in \mathcal{U}$.

Integrating in the common offset u turns the sum for each ball into an integral over the disjoint half-open cells

$$a_B + (N^2 z', j) + \mathcal{U}, \quad |z'| \leq 1, \quad 0 \leq j < N^2.$$

These cells cover B . Indeed, for $y \in B$ set $r = y - a_B$; then

$$|r'| \leq R \leq N^2/4, \quad N^2/4 \leq r_3 \leq 3N^2/4.$$

The lower corner ℓ' of its horizontal cell in $D^{-1}\mathbb{Z}^2$ satisfies $|\ell'| \leq N^2/4 + \sqrt{2}/D < N^2$ for the fixed large N under consideration. Hence $z' = \ell'/N^2$ is a retained label, and $j = \lfloor r_3 \rfloor$ is one of the retained times. The change of variables has Jacobian one; the offset domain has fixed volume D^{-2} . Equation (5.14) therefore implies

$$\sum_{B \in \mathcal{B}} \int_B |F_*(y)|^3 dy \leq CP(\kappa, \eta) M(s)^{3/2} N^{\gamma_0} \log^2(2 + N).$$

On $\{|F| > \mu\}$, Equation (5.3) gives $|F_*| > \mu/2$. Chebyshev's inequality now proves Equation (5.1).

All choices made in this proof involve only fixed geometric constants and fixed orders of differentiation. Thus μ_0 is independent of κ, η, s , as asserted. $\square$

6 Global bounds and exponent dependence

To apply Proposition 5.1 globally, we cover each superlevel set by a controlled number of sparse families. The L^4 and gradient bounds below control the number of occupied unit cubes; slow growth of local point counts then selects the radii without a diameter bound.

We retain the fixed graph, support, and window from the preceding sections. All constants in this section, except for the displayed dependence on the small exponents, depend only on that fixed setup. The elementary estimates below use only Equation (2.1). The global estimate for $3 < p \leq 4$ additionally uses Proposition 5.1, and hence the packet input in Theorem 3.4.

6.1 Elementary bounds and the number of occupied cubes

Lemma 6.1 (Basic graph estimates). *Let $g \in L^2(\mathbb{R}^2)$ have compact support, and let*

$$G(y', t) = \int_{\mathbb{R}^2} g(\xi) e(y' \cdot \xi + t\phi(\xi)) d\xi.$$

If $\phi'' \geq cI$ on $\mathbb{R}^2$, then

$$\|G\|_{L^4(\mathbb{R}^3)} \leq (\pi/c)^{1/4} \|g\|_{L^2(\mathbb{R}^2)}. \quad (6.1)$$

In particular, for $F = E_\phi(h\chi_1^{\nu*})$ with h supported in Q_0 , there is a fixed $C_b \geq 1$ such that

$$\|F\|_4 + \|F\|_\infty + \|\nabla F\|_\infty \leq C_b \|h\|_3. \quad (6.2)$$

Proof. Let λ be the finite complex measure on the graph of ϕ whose density in base coordinates is g . For $v \in C_c^\infty(\mathbb{R}^3)$, Cauchy–Schwarz gives

$$\left| \int v d(\lambda * \lambda) \right| \leq \|g\|_2^2 \left(\iint_{\mathbb{R}^2 \times \mathbb{R}^2} |v(\xi + \xi', \phi(\xi) + \phi(\xi'))|^2 d\xi d\xi' \right)^{1/2}.$$

Put $m = \xi + \xi'$ and $w = (\xi - \xi')/2$; this linear change has absolute Jacobian one. Write $w = r\omega$, where $r > 0$ and $\omega \in S^1$, and set

$$H_{m,\omega}(r) = \phi(m/2 + r\omega) + \phi(m/2 - r\omega).$$

Uniform convexity implies

$$H'_{m,\omega}(r) = \int_{-r}^r \omega^\top \phi''(m/2 + s\omega) \omega ds \geq 2cr.$$

Thus $H_{m,\omega}$ is strictly increasing for $r > 0$, and the substitution $s = H_{m,\omega}(r)$ yields

$$\begin{aligned} & \iint |v(\xi + \xi', \phi(\xi) + \phi(\xi'))|^2 d\xi d\xi' \\ &= \int_{\mathbb{R}^2} \int_{S^1} \int_0^\infty |v(m, H_{m,\omega}(r))|^2 r dr d\omega dm \leq \frac{\pi}{c} \|v\|_2^2. \end{aligned}$$

The omitted set $r = 0$ has measure zero. Consequently $\lambda * \lambda$ has an L^2 density with norm at most $(\pi/c)^{1/2} \|g\|_2^2$, by the L^2 representation theorem. Its Fourier transform is $G(-\cdot)^2$ in the sense of tempered distributions. Plancherel therefore gives Equation (6.1); this reasoning applies to the stated rough data and does not assume the convolution has a density in advance.

For $g = h\chi_1^{\nu*}$, compact support and Hölder's inequality bound $\|g\|_2$ by a fixed multiple of $\|h\|_3$. Furthermore

$$\|F\|_\infty \leq \|g\|_1, \quad \|\nabla F\|_\infty \leq 2\pi \int |(\xi, \phi(\xi))| |g(\xi)| d\xi.$$

Differentiation under the integral is valid because the integrands are supported in a fixed compact set. Another application of Hölder's inequality proves Equation (6.2). $\square$

For the covering argument, normalize $\|h\|_3 \leq 1$ and write

$$A_\mu = \{y \in \mathbb{R}^3 : |F(y)| > \mu\}, \quad 0 < \mu \leq 1.$$

The half-open cubes $k + [0, 1)^3$, $k \in \mathbb{Z}^3$, form a partition. Call such a cube occupied if it meets A_μ .

Lemma 6.2 (Finite occupancy). *There is a fixed $C_X \geq 1$ such that at most $C_X \mu^{-7}$ unit cubes are occupied. In particular, if $\mu \leq C_X^{-1}$, one can choose one point of A_μ in each occupied cube to obtain a finite set X satisfying $\#X \leq \mu^{-8}$.*

Proof. First take any finite collection $\mathcal{Q}$ of occupied cubes and choose $x_Q \in Q \cap A_\mu$ for each of them. By Equation (6.2), on $B(x_Q, r_\mu)$, where $r_\mu = \mu/(2C_b) \leq 1/2$, one has $|F| > \mu/2$. These balls have overlap at most 125: if a point belongs to one of them, its center lies within distance one, and at most 5^3 unit lattice cubes can contain such centers. Separation of the selected points is not required. Therefore

$$\#\mathcal{Q} \frac{\pi}{96C_b^3} \mu^7 \leq \sum_{Q \in \mathcal{Q}} \int_{B(x_Q, r_\mu)} |F|^4 \leq 125 \|F\|_4^4 \leq 125 C_b^4.$$

For example, one may take $C_X = \max\{1, 12000C_b^7/\pi\}$. This bound holds for every finite collection of occupied cubes. An infinite collection would have finite subsets of arbitrarily large cardinality, a contradiction. Thus the full collection is finite with the same bound. Finally $C_X \mu^{-7} \leq \mu^{-8}$ when $\mu \leq C_X^{-1}$. $\square$

6.2 A multiscale cover without a diameter bound

The finite-growth and coloring argument below is the covering step of the sparse-ball method; compare Bourgain–Guth [3, arXiv version 3, Appendix, Lemma A.2]. We retain explicit scale choices to track the dependence on the output exponent.

Fix once and for all a number $\mu_0 > 0$ that is no larger than the threshold in Proposition 5.1, C_X^{-1} , and $1/2$. In particular, for $0 < \mu \leq \mu_0$ we have

$$\mu^{-60} \geq 8, \quad \mu^{-100} > 2. \quad (6.3)$$

This choice is independent of all small exponents below.

Lemma 6.3 (Sparse covering). *Let $0 < \tau \leq 1$, $0 < \mu \leq \mu_0$, and let X be the set in Lemma 6.2. Set*

$$\beta = \frac{\tau}{4}, \quad J = \left\lceil \frac{32}{\tau} \right\rceil + 1, \quad R_j = \mu^{-100(j+1)} \quad (0 \leq j \leq J). \quad (6.4)$$

The union of the occupied cubes is covered by at most

$$3J(1 + 8 \log_2(1/\mu)) \mu^{-\beta} \quad (6.5)$$

families of balls. Each family has at most μ^{-8} balls, all of radius $R = 4R_j$ for some $0 \leq j < J$, and its centers are separated by at least $\mu^{-40}R$. For the dyadic N chosen in Proposition 5.1,

$$4R \leq N^2 < 16R, \quad N < 8\mu^{-50J}. \quad (6.6)$$

Proof. If X is empty there is nothing to cover. Otherwise, using open balls throughout, set

$$n_j(x) = \#(X \cap B(x, R_j)), \quad x \in X.$$

For each x there is an index $0 \leq j < J$ with

$$n_{j+1}(x) \leq \mu^{-\beta} n_j(x). \quad (6.7)$$

Indeed, failure at every index would imply $n_J(x) > \mu^{-\beta J} n_0(x) \geq \mu^{-\beta J} > \mu^{-8}$, since $\beta J > 8$, contrary to $\#X \leq \mu^{-8}$. Assign each x one such index j and the unique integer $k \geq 0$ for which

$$2^k \leq n_j(x) < 2^{k+1}.$$

There are at most $J(1 + 8 \log_2(1/\mu))$ nonempty groups with a common pair (j, k) .

In one group, choose a maximal pairwise disjoint collection of the open balls $B(x, R_j)$, and denote its centers by Y . The collection is finite. Each unselected ball intersects a selected one, so every center in the group is within distance $2R_j$ of a point of Y . A point of the unit cube containing that center is within a further distance $\sqrt{3}$. Since $R_j \geq \mu^{-100}$, the balls $\{B(y, 4R_j) : y \in Y\}$ cover all cubes assigned to the group.

Make a finite graph on Y , joining distinct x, y when $|x - y| < R_{j+1}/2$. For a fixed vertex x , the original balls $B(y, R_j)$ of all its neighbors are disjoint and lie in $B(x, R_{j+1})$: Equation (6.3) gives $R_j < R_{j+1}/2$. Each neighbor ball contains at least 2^k points of X , whereas Equation (6.7) gives

$$n_{j+1}(x) < \mu^{-\beta} 2^{k+1}.$$

The degree of x is therefore less than $2\mu^{-\beta}$. A greedy coloring uses at most $3\mu^{-\beta}$ colors. Centers of the same color are separated by at least $R_{j+1}/2$, and

$$\frac{R_{j+1}}{2} = \frac{\mu^{-100}}{2} R_j \geq \mu^{-40} (4R_j)$$

by Equation (6.3). Each color thus supplies a family satisfying the separation condition, with at most $\#X \leq \mu^{-8}$ balls and $R = 4R_j \geq \mu^{-100}$.

Multiplying the group and color counts proves Equation (6.5). A dyadic N with $4R \leq N^2 < 16R$ exists, and satisfies

$$N < 4\sqrt{R} = 8\mu^{-50(j+1)} \leq 8\mu^{-50J}.$$

The argument uses neither the absolute locations of the points nor the diameter of X . In particular, different clusters may be arbitrarily far apart. The use of open balls also resolves boundary cases: unselected balls meet selected balls by maximality, and equality in the graph's distance threshold leaves the desired weak separation inequality intact. $\square$

6.3 Integration of the distribution bound

Theorem 6.4 (Global graph estimate). *Fix a phase and window setup for which Theorem 3.4 and Proposition 5.1 apply. For $3 < p \leq 4$, put*

$$\tau = p - 3, \quad \theta = 10^{-6}\tau^2. \quad (6.8)$$

There is a constant C , independent of p in this interval, such that

$$\|E_\phi(h\chi_1^{\nu*})\|_p \leq C \left[P(\theta, \theta) M(\theta)^{3/2} \tau^{-7} \right]^{1/p} \|h\|_3 \quad (6.9)$$

for all $h \in L^3$ supported in Q_0 . For $4 < p < \infty$, Equation (6.2) alone gives $\|E_\phi(h\chi_1^{\nu})\|_p \leq C_b \|h\|_3$.*

Proof. By homogeneity it suffices to take $\|h\|_3 \leq 1$. In Proposition 5.1 choose $\kappa = \eta = s = \theta$. These choices lie in the permitted parameter ranges, and

$$\gamma_0 = 10\kappa + \eta + \frac{3}{2}s = \frac{25}{2}\theta.$$

For J in Equation (6.4), we have

$$J \leq \frac{34}{\tau}, \quad 50J\gamma_0 \leq \frac{17}{800}\tau < \frac{\tau}{4}. \quad (6.10)$$

Consequently the scale in every family furnished by Lemma 6.3 satisfies

$$N^{\gamma_0} \leq 8^{\gamma_0} \mu^{-\tau/4} \leq 2\mu^{-\tau/4}, \quad \log(2 + N) \leq 53J(1 + \log(1/\mu)).$$

Applying Proposition 5.1 to each family and summing over the cover yields, for $0 < \mu \leq \mu_0$,

$$|A_\mu| \leq CP(\theta, \theta)M(\theta)^{3/2}J^3(1 + \log(1/\mu))^3\mu^{-3-\tau/2}. \quad (6.11)$$

Here the colors cost $\mu^{-\beta} = \mu^{-\tau/4}$, while the scale factor costs at most $\mu^{-\tau/4}$. The group count contributes one factor of $J(1 + \log(1/\mu))$, and the logarithm squared in Proposition 5.1 contributes the other two. All constants and μ_0 are independent of τ .

The layer-cake identity, valid by Tonelli even before finiteness of the norm is known, is

$$\|F\|_p^p = p \int_0^\infty \mu^{p-1} |A_\mu| d\mu.$$

For its portion below μ_0 , Equation (6.11) and $p = 3 + \tau$ reduce the needed integral to

$$\int_0^1 \mu^{-1+\tau/2} (1 + \log(1/\mu))^3 d\mu = \frac{2}{\tau} + \frac{12}{\tau^2} + \frac{48}{\tau^3} + \frac{96}{\tau^4} \leq 158\tau^{-4}. \quad (6.12)$$

For example, the substitution $u = \log(1/\mu)$ turns the integral into $\int_0^\infty e^{-\tau u/2} (1+u)^3 du$, which gives the displayed identity by integrating each monomial. Since $p \leq 4$ and $J^3 \leq 34^3\tau^{-3}$, the small-level contribution is at most $CP(\theta, \theta)M(\theta)^{3/2}\tau^{-7}$.

For the remaining levels, again by Tonelli,

$$p \int_{\mu_0}^\infty \mu^{p-1} |A_\mu| d\mu \leq \int_{\{|F| > \mu_0\}} |F|^p \leq \mu_0^{p-4} \|F\|_4^4 \leq \mu_0^{-1} C_b^4.$$

This is uniform for $3 < p \leq 4$ and is absorbed into the previous bound because $P, M \geq 1$ and $\tau \leq 1$. Taking the p th root and restoring homogeneity proves Equation (6.9). If $p > 4$, directly use

$$\|F\|_p^p \leq \|F\|_\infty^{p-4} \|F\|_4^4 \leq C_b^p \|h\|_3^p.$$

□

The factor τ^{-7} explicitly records the elementary loss removal argument. The assumptions provide finite constants $P(\theta, \theta)$ and $M(\theta)$, but do not provide rates for them as $\theta \downarrow 0$. Equation (6.9) therefore asserts no polynomial upper bound in $(p-3)^{-1}$ for the full extension constant.

6.4 A necessary divergence at the endpoint

Proposition 6.5 (Lower bound for the best constant). *For every nonempty compact smooth surface Σ in the stated class there is $c_\Sigma > 0$ such that its best extension constant satisfies*

$$C_{\Sigma, p}^{\text{best}} \geq c_\Sigma (p-3)^{-1/p}, \quad 3 < p \leq 4. \quad (6.13)$$

This conclusion does not use the Kakeya or packet estimates.

Proof. Choose a nonempty interior chart; a nonempty smooth surface with boundary also has such a chart. Write it in the affine graph form

$$\Psi(\xi) = v_0 + L(\xi, \phi(\xi)), \quad \xi \in D,$$

where L is invertible, and write $d\sigma = J_\Sigma(\xi) d\xi$ on this chart. Take a fixed nonzero $g \in C_c^\infty(D)$ and define $f(\Psi(\xi)) = g(\xi)/J_\Sigma(\xi)$ on the chart, with $f = 0$ elsewhere. This is a smooth surface datum with

$$B_f := \sup_{3 \leq p \leq 4} \|f\|_{L^p(\Sigma)} < \infty.$$

Let

$$G(y', t) = \int g(\xi) e(y' \cdot \xi + t\phi(\xi)) d\xi, \quad A = \|g\|_2^2 > 0.$$

Horizontal Plancherel and its derivative identity give, for $t \geq 0$,

$$\begin{aligned} \int_{\mathbb{R}^2} |G(y', t)|^2 dy' &= A, \\ \int_{\mathbb{R}^2} |y'|^2 |G(y', t)|^2 dy' &= \frac{1}{(2\pi)^2} \|\nabla(g e(t\phi))\|_2^2 \leq B_g(1+t)^2, \end{aligned}$$

where one may take $B_g = 2(2\pi)^{-2} \|\nabla g\|_2^2 + 2\|g \nabla \phi\|_2^2 > 0$. Set $K = \max\{1, (2B_g/A)^{1/2}\}$. The integral outside the disk $|y'| < K(1+t)$ is at most $B_g/K^2 \leq A/2$, so at least $A/2$ of the squared norm lies inside that disk. Hölder's inequality on the disk implies

$$\int_{\mathbb{R}^2} |G(y', t)|^p dy' \geq (A/2)^{p/2} (\pi K^2)^{1-p/2} (1+t)^{2-p} \geq c_g (1+t)^{2-p}, \quad (6.14)$$

uniformly for $3 \leq p \leq 4$, with the explicit positive choice

$$c_g = \min \left\{ (A/2)^{3/2} (\pi K^2)^{-1/2}, (A/2)^2 (\pi K^2)^{-1} \right\}.$$

Integrating Equation (6.14) over $t \geq 0$ yields $\|G\|_p^p \geq c_g/(p-3)$.

The affine coordinates introduce exactly the factors

$$E_\Sigma f(x) = e(x \cdot v_0) G(L^\top x), \quad \|E_\Sigma f\|_p^p = |\det L|^{-1} \|G\|_p^p.$$

Hence, on setting $d_g = |\det L|^{-1} c_g > 0$, we obtain

$$C_{\Sigma, p}^{\text{best}} \geq B_f^{-1} d_g^{1/p} (p-3)^{-1/p} \geq B_f^{-1} \min\{d_g^{1/3}, d_g^{1/4}\} (p-3)^{-1/p}.$$

This proves Equation (6.13), with every choice of chart and datum fixed independently of p . $\square$

7 Surface estimates and consequences

7.1 Passage to the surface

We now deduce Theorem 1.1 from Theorem 6.4. Cover Σ by finitely many sufficiently small charts. By Lemma 3.1, after fixed affine normalizations each chart lies in a graph with a phase admissible for Theorem 3.4. The construction and its constants are fixed separately for each chart and do not depend on p . A measurable partition subordinate to the covering avoids any assumption that the data extend smoothly across chart boundaries.

Write one chart in ambient frequency coordinates as

$$\xi = v_0 + L(\zeta, \phi(\zeta)), \quad \zeta \in A \subset \mathbb{R}^2,$$

where L is invertible and A is bounded, possibly cut off by the boundary of the surface. Its surface measure is $J_\Sigma(\zeta) d\zeta$, with J_Σ smooth and strictly positive on a neighborhood of the chart. For data supported on this piece, set

$$g(\zeta) = \begin{cases} f(v_0 + L(\zeta, \phi(\zeta))) J_\Sigma(\zeta), & \zeta \in A, \\ 0, & \zeta \notin A. \end{cases}$$

Then

$$E_\Sigma f(x) = e(x \cdot v_0) \int_{\mathbb{R}^2} g(\zeta) e((L^\top x)' \cdot \zeta + (L^\top x)_3 \phi(\zeta)) d\zeta.$$

The change of physical variables $y = L^\top x$ contributes $|\det L|^{-1/p}$ to the L^p norm. Moreover,

$$\begin{aligned} \|g\|_3^3 &\leq (\sup_A J_\Sigma)^2 \int_{\Sigma_{\text{chart}}} |f|^3 d\sigma, \\ \|f\|_{L^3(\Sigma_{\text{chart}})} &\leq \sigma(\Sigma_{\text{chart}})^{1/3-1/p} \|f\|_{L^p(\Sigma_{\text{chart}})}, \quad p > 3. \end{aligned}$$

After the finite window partition of Equation (2.2), Theorem 6.4 applies to every resulting term. The determinant, area, window, and chart factors are bounded uniformly for $3 < p \leq 4$. Summing the finitely many terms by the triangle inequality gives Equation (1.1) in this range, with the quantitative bound Equation (1.3). Here P_Σ is the maximum of the finitely many packet constants, enlarged to be at least one.

For $p > 4$, the local L^4 and L^∞ bounds give the result directly, as in Theorem 6.4, and the same finite chart argument applies. No smoothness of f , nor of its extension by zero on the parameter plane, has been used. Boundary charts are therefore included.

Proposition 6.5 supplies Equation (1.4) for every nonempty surface by choosing a nonzero smooth datum in an interior chart. This completes the proof of Theorem 1.1 and the stated dependence on p .

7.2 Free Schrödinger local smoothing

The extension estimate for a compact paraboloid controls all frequencies of the Schrödinger propagator through the transfer theorem of Lee, Rogers, and Seeger. The strict Sobolev inequality in Corollary 1.2 then supplies the summability across frequency scales.

Proof of Corollary 1.2. On the compact paraboloid $\Sigma = \{(y, |y|^2) : |y| \leq 1\}$, write $d\sigma = w(y) dy$, where $w(y) = \sqrt{1 + 4|y|^2}$. Apply Theorem 1.1 to the datum $g(y)/w(y)$ on Σ . A fixed change of output variables then gives

$$\left\| \int_{|y| \leq 1} g(y) e^{i(\tau|y|^2 - \xi \cdot y)} dy \right\|_{L^p(\mathbb{R}_{\xi, \tau}^3)} \leq C_p \|g\|_{L^p(\{|y| \leq 1\})}.$$

This is the compact-frequency extension estimate in Lee–Rogers–Seeger [10, Equation (1.3)] with both exponents equal to p . Their Theorem 1.1, with spatial dimension two and Besov summation exponent two, yields

$$\|e^{it\Delta} f\|_{L^p(\mathbb{R}^2 \times [0,1])} \leq C_p \|f\|_{B_{\alpha,2}^p(\mathbb{R}^2)}, \quad \alpha = 2(1 - 2/p) - 2/p = 2 - 6/p.$$

For an inhomogeneous Littlewood–Paley partition $(P_k)_{k \geq 0}$, the uniform annular multiplier bound $\|P_k f\|_p \leq C_s 2^{-ks} \|(I - \Delta)^{s/2} f\|_p$ gives

$$\|f\|_{B_{\alpha,2}^p} \leq C_s \left(\sum_{k \geq 0} 2^{-2k(s-\alpha)} \right)^{1/2} \|f\|_{W^{s,p}}.$$

The series converges for $s > \alpha$. This proves Equation (1.2) on Schwartz functions, and density gives the stated bounded extension. $\square$

7.3 Translation-invariant oscillatory integrals

A second application concerns phases whose curvature changes only by a scalar factor in time. Such a phase can be straightened by changing the output variables alone, reducing the estimate to one fixed elliptic graph. This is the translation-invariant setting of the output straightening theorem of Gao–Liu–Xi [5, Theorem 1.12 and Section 4.1]. We give the short argument for the two-dimensional input considered here, then allow a general smooth amplitude by Fourier expansion.

Corollary 7.1 (Elliptic translation-invariant Bourgain phases). *Let*

$$\phi(x, t; y) = x \cdot y + \psi(t; y), \quad x, y \in \mathbb{R}^2, \quad t \in \mathbb{R}, \quad \psi(0; y) = 0,$$

be a fixed real smooth phase near the origin. Suppose throughout this neighborhood that

$$M(t, y) = D_y^2 \partial_t \psi(t; y) \quad \text{is definite}, \quad \partial_t M(t, y) = c(t, y) M(t, y)$$

for a smooth real scalar function c . On a sufficiently small fixed product chart, let a be any fixed smooth amplitude with compact support in its interior. For every $3 < p < \infty$,

$$\left\| \int e^{iN\phi(x, t; y)} a(x, t; y) f(y) dy \right\|_{L^p(\mathbb{R}_{x, t}^3)} \leq C_{\phi, a, p} N^{-3/p} \|f\|_{L^p(\mathbb{R}^2)}, \quad N \geq 1, \quad (7.1)$$

where the operator is zero outside the output support of a . The constant may depend on the fixed chart, phase, amplitude and p , but not on N or f .

Proof. Shrink the product neighborhood so that its input domain is a ball. Since M is a Hessian in y , commuting its derivatives in $\partial_t M = cM$ gives

$$(\partial_{y_k} c) M_{ij} = (\partial_{y_i} c) M_{kj}.$$

Multiplication by $(M^{-1})_{j\ell}$, summed over j , yields $(\partial_{y_k} c) \delta_{i\ell} = (\partial_{y_i} c) \delta_{k\ell}$. For each k , choose $\ell = i \neq k$; hence $\nabla_y c = 0$. The input ball is connected, so $c = c(t)$. Set

$$\rho(t) = \exp\left(\int_0^t c(v) dv\right), \quad h(y) = \partial_t \psi(0; y), \quad S(t) = \int_0^t \rho(v) dv.$$

The equation for M now implies $D_y^2(\partial_t \psi - \rho h) = 0$. Thus $\partial_t \psi = \rho(t)h(y) + b(t) \cdot y + d(t)$ for smooth functions b, d . Integrating and using $\psi(0; y) = 0$ gives

$$\psi(t; y) = S(t)h(y) + B(t) \cdot y + D(t), \quad B(t) = \int_0^t b(v) dv, \quad D(t) = \int_0^t d(v) dv.$$

Since $S' = \rho > 0$, it has a smooth local inverse $t = t(s)$. The output change $\kappa(u, s) = (u - B(t(s)), t(s))$ therefore satisfies

$$\phi(\kappa(u, s); y) = u \cdot y + sh(y) + r(s), \quad r(s) = D(t(s)).$$

Here $D_y^2 h = M(0, y)$ is definite. No input coordinate change is needed.

Put $z = (u, s)$ and $A(z, y) = a(\kappa(z); y)$. Extend this fixed amplitude smoothly by zero, and periodically in z across a fixed cube Q of side L whose interior contains its output support. On Q ,

$$A(z, y) = \sum_{k \in \mathbb{Z}^3} e^{2\pi i k \cdot z / L} A_k(y), \quad \sum_k \|A_k\|_\infty < \infty.$$

Indeed, integration by parts in z gives $\|A_k\|_\infty \leq C_m(1 + |k|)^{-m}$ for any fixed $m > 3$. All coefficients have a common compact frequency support, and the constants are independent of N . Choose a compact graph patch Σ , parametrized by $\omega(y) = (y, h(y))$, with smooth boundary and parameter domain containing this support, and write $d\sigma = w(y) dy$, where $w(y) = \sqrt{1 + |\nabla h(y)|^2}$. For

$$g_k(\omega(y)) = A_k(y)f(y)/w(y)$$

we have

$$\|g_k\|_{L^p(\Sigma)}^p = \int |A_k f|^p w^{1-p} dy \leq \|A_k\|_\infty^p \|f\|_p^p.$$

The factor $e^{iNr(s)}$ is unimodular, and each remaining Fourier term is $e^{2\pi i k \cdot z/L} E_\Sigma g_k(Nz/(2\pi))$. Minkowski's inequality and Theorem 1.1, applied globally to each term, bound their sum on Q by

$$(2\pi/N)^{3/p} C_{\Sigma,p} \sum_k \|A_k\|_\infty \|f\|_p.$$

Changing back by κ contributes only the bounded factor $\sup |\det D\kappa|^{1/p}$ on the fixed output support. Absolute coefficient summability and the local L^1 integrability of f justify the series and its interchange with the integral. This proves Equation (7.1), with no power loss in N . $\square$

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